SOFTWARE ENGINEERING (LAB-I)

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SECTION: 5TH CSE 'D'

Requirements Engineering & UML Use-Case Modelling

TOPIC: Patient Health Record Consent Management System

Introduction and Scenario review:

Use case (UC-01)

Preconditions: the patient is registered used of the system and atleast on clinic or provider available to grant the consent to.

Post conditions: an active consent record exists, linking the patient, the provider, the approved data categories and expiry date; the event is written in audit log.

Main success catalogue:

1. Patient select 'manage consent' from the home screen. System prompts for authentication (<<include>> UC-01 authentic user)
2. Patient completes multi-factor authentication and is directed to the consent dashboard
3. Patient selects "Grant New Consent", system displays a form listing available clinics/providers and record categories (e.g: visit history, prescriptions, lab results).
4. Patient selects a provider, checks the desired data categories and sets an expiry date
5. Patient reviews a summary screen showing exactly what will be shared and confirms
6. System validates the input (provider exists (success))
7. System creates a consent record with status "Active" ad writes an entry to audit log
8. System sends a confirmation message "Consent granted to [provider] until [date]"
9. The new consent appears at the top of the patient's consent list

10. Use case ends successfully

Alternate flow:

a) Validation Fails:
   - System displays an inline error (“Expiry date must be in the future” or “select atleast one category”)
   - Patient corrects the form and resubmits
   - If the patient cancels instead, no consent record is created and the use case ends without effect

b) Authentication fails:
   - System displays “Authentication Failed” and returns to the login page
   - After three failed attempts the account would be temporarily locked and patient is notified by email
   - Use case ends without granting consent

Elicit and Draft Requirements:

1. Elicited Functional Requirements

ID	Functional Requirement
FR-01	The system shall allow a registered patient to access Manage Consent from the home screen.
FR-02	The system shall authenticate the patient before allowing consent management.
FR-03	The system shall support multi-factor authentication (MFA) for patient authentication.
FR-04	The system shall display a consent dashboard after successful authentication.
FR-05	The system shall allow the patient to select Grant New Consent.
FR-06	The system shall display available clinics/providers to whom consent can be granted.
FR-07	The system shall display selectable healthcare data categories such as visit history, prescriptions, and lab results.
FR-08	The system shall allow the patient to select one provider and one or more data categories.
FR-09	The system shall allow the patient to specify an expiry date for the consent.
FR-10	The system shall display a summary of the provider, selected data categories, and expiry date before confirmation.
FR-11	The system shall validate that the selected provider exists and is available.
FR-12	The system shall validate that at least one data category has been selected.
FR-13	The system shall validate that the expiry date is in the future.
FR-14	The system shall create an Active Consent record when all validations succeed.
FR-15	The consent record shall link the patient, provider, approved data categories, and expiry date.
FR-16	The system shall record the consent-granting event in the audit log.
FR-17	The system shall send a confirmation notification to the patient after consent is successfully granted.
FR-18	The system shall display the newly created consent at the top of the patient's consent list.
FR-19	The system shall allow the patient to cancel before confirmation without creating a consent record.
FR-20	The system shall temporarily lock the account after three consecutive authentication failures.
FR-21	The system shall notify the patient by email when the account is temporarily locked.

2. Non-Functional Requirements

ID	Category	Requirement
NFR-01	Security	Patient authentication shall use MFA before consent can be granted.
NFR-02	Security	Consent information and patient data shall be protected against unauthorized access.
NFR-03	Security	Only authorized patients shall be able to create or modify their consent records.
NFR-04	Auditability	Every successful consent-granting operation shall be recorded in an immutable audit log.
NFR-05	Privacy	The system shall share only the data categories explicitly approved by the patient.
NFR-06	Accuracy	The system shall accurately associate the consent with the selected patient and provider.
NFR-07	Usability	Validation errors shall be displayed inline and clearly explain how the patient can correct them.
NFR-08	Availability	Consent management should be available whenever the healthcare system is operational.
NFR-09	Performance	Consent confirmation and record creation should complete within about 2–3 seconds under normal load.

Draft Requirements

You can convert the elicited requirements into a more formal **Software Requirements Specification (SRS)** format.

3.1 Patient Authentication

DR-01: The system shall require a registered patient to authenticate before accessing the consent management functionality.

DR-02: The system shall require successful MFA before displaying the consent dashboard.

DR-03: If authentication fails, the system shall display an **"Authentication Failed"** message and redirect the patient to the login page.

DR-04: After three consecutive failed authentication attempts, the system shall temporarily lock the patient's account and notify the patient through email.

3.2 Consent Creation

DR-05: The system shall provide a **Grant New Consent** option on the consent dashboard.

DR-06: The system shall display all eligible clinics/providers available for consent.

DR-07: The system shall allow the patient to select the provider to whom access will be granted.

DR-08: The system shall display available healthcare data categories, including but not limited to:

- Visit history
- Prescriptions
- Lab results
- Medical records

DR-09: The system shall allow the patient to select one or more data categories.

DR-10: The system shall allow the patient to specify an expiry date for the consent.

3.3 Validation

DR-11: The system shall verify that the selected provider exists and is eligible to receive the requested data.

DR-12: The system shall reject the request if no data category is selected.

DR-13: The system shall reject an expiry date that is equal to or earlier than the current date.

DR-14: The system shall display an appropriate inline error message when validation fails.

DR-15: The patient shall be able to correct the invalid information and resubmit the consent request.

3.4 Consent Confirmation

DR-16: Before creating the consent record, the system shall display a confirmation summary containing:

Patient → Provider → Data Categories → Expiry Date

DR-17: The system shall create the consent only after the patient explicitly confirms the summary.

DR-18: If the patient cancels before confirmation, the system shall terminate the operation without creating a consent record.

3.5 Consent Record

DR-19: Upon successful confirmation and validation, the system shall create a consent record with status **Active**.

The record shall contain at minimum:

Consent ID, Patient ID, Provider ID, Approved Data Categories, Creation Date, Expiry Date

,Status

DR-20: The system shall associate the consent record with exactly one patient and the selected provider.

DR-21: The system shall ensure that only the data categories explicitly approved by the patient are covered by the consent.

3.6 Audit and Notification

DR-22: The system shall create an audit-log entry whenever a new consent is successfully granted.

DR-23: The system shall send the patient a confirmation message after successful consent creation.

DR-24: The newly created consent shall appear at the top of the patient's consent list.

Actors

Actor	Role in the System
1. Patient	Registers/logs in, authenticates using MFA, grants consent, selects providers and data categories, and manages consent.
2. Healthcare Provider / Clinic	Receives access to the patient's approved healthcare data based on an active consent.
3. Authentication Service	Verifies the patient's credentials and MFA during authentication.
4. Notification Service	Sends confirmation messages and account-lock notifications to the patient.
5. Audit Log System	Records consent-related events for security and traceability.

Use Cases

ID	Use Case	Description
UC-01	**Grant New Consent**	Patient grants a provider access to selected healthcare data for a specified period.
UC-02	**Authenticate User**	Patient authenticates using credentials and MFA before accessing consent management.
UC-03	**View Consent Dashboard**	Patient views existing active/expired consent records.
UC-04	**Manage Consent**	Patient manages their existing consent records.
UC-05	**Validate Consent Request**	System verifies provider, data categories, and expiry date.

ID	Use Case	Description
UC-06	**Record Consent in Audit Log**	System records the successful consent operation for traceability.

USE-CASE DIAGRAM:

